

Causal Representation Learning and Causal Discovery in Deep Learning

Abstract

Causal representation learning and causal discovery have become two of the most active interfaces between causality and modern deep learning. The former asks how to recover latent variables with causal semantics from high-dimensional observations such as images, video, audio, and multimodal streams; the latter asks how to recover causal structure, or at least a Markov-equivalence class or posterior over structures, from observational, interventional, temporal, or weakly supervised data. These agendas are complementary. Causal discovery traditionally presumes that the relevant variables are already given, while deep learning is most useful precisely when the right variables are unknown and must be learned from raw data. Conversely, representation learning by itself is often statistically underdetermined, and causality supplies invariance principles, mechanism modularity, identifiability conditions, and intervention semantics that can constrain the latent space in meaningful ways [1–12]. Recent surveys and foundational position papers converge on this diagnosis and frame causal representation learning as a central open problem for AI, while causal discovery has simultaneously evolved from classical conditional-independence and score-based algorithms to differentiable, Bayesian, and generative methods that leverage neural parameterizations and continuous optimization. ¹

This survey reviews the field with a focus on deep-learning-centered methodology. It first situates the area in the structural causal model framework and classical identifiability results. It then organizes methods into major families: classical discovery foundations and their neural descendants; differentiable and Bayesian graph learning; temporal and interventional discovery; identifiable nonlinear independent component analysis and related theories of latent-variable recovery; weakly supervised, temporal, multi-view, and multi-environment causal representation learning; and deep structural causal models and causally informed generative models. Throughout, the survey emphasizes recurring themes: identifiability versus expressivity, interventions versus auxiliary structure, equivariance and invariance as substitutes for direct supervision, uncertainty quantification, benchmark design, and the tension between synthetic rigor and realistic high-dimensional data. The paper concludes by arguing that future progress will likely depend on tighter integration of discovery, representation, and downstream causal reasoning, together with stronger evaluation protocols and more realistic yet controllable benchmarks [6–12].

Introduction

The central motivation behind causal approaches to deep learning is that predictive success alone does not guarantee invariance under intervention, distribution shift, or compositional reuse. Standard deep models can compress high-dimensional observations into useful latent states, but the meaning of these states is usually entangled, non-identifiable, and unstable across environments. Classical causality, by contrast, offers an explicit language for interventions, counterfactuals, modular mechanisms, and transport across changing domains. The difficulty is that most causal theory assumes that the causal variables are already observed, while most modern machine learning problems begin with unstructured observations from which the variables themselves must be inferred. Causal representation learning is therefore the problem of discovering or identifying latent variables that are not merely useful summaries,

but variables with some causal semantics, ideally admitting manipulation, modularity, and transportability [1, 4–7]. This framing has become canonical in recent work and is now widely treated as the bridge between graphical causality and representation learning. ²

A second motivation is that structure discovery itself has changed in the deep-learning era. Traditional causal discovery relied on conditional-independence testing, combinatorial search, or restricted functional models. That line of work remains foundational, but it has been joined by neural score models, differentiable acyclicity penalties, variational posteriors over graphs, GFlowNet and diffusion-based search, and hybrid systems that perform discovery and inference jointly [8–10, 28–46]. These developments were driven by at least four pressures. First, nonlinear relations are common in scientific, economic, and biological systems. Second, modern datasets often mix continuous, categorical, temporal, and missing data. Third, many use cases require posterior uncertainty over graphs rather than a single point estimate. Fourth, high-dimensional domains such as vision and robotics force a merger between representation learning and discovery [8–12, 28–46].

The survey adopts the structural causal model as its basic conceptual framework. An SCM specifies endogenous variables, exogenous noise variables, and structural assignments that determine each variable from its parents and an exogenous disturbance [1, 4, 5]. In causal discovery, the objective is to recover the graph, functional family, or equivalence class from data. In causal representation learning, the objective is to recover latent variables that support an SCM over the latent space, together with an encoder-decoder or generative process linking those latents to high-dimensional observations. This distinction matters because discovery over observed tabular variables can sometimes be identifiable under classical assumptions, whereas latent-variable recovery is generically impossible in the absence of additional supervision or inductive bias. Consequently, nearly all practical CRL methods rely on extra structure: nonstationarity, multiple environments, sparse interventions, temporality, partial labels, equivariances, multiple views, or architectural factorization [6, 11, 61–89].

Another conceptual thread running through the literature is the independent causal mechanisms principle. Informally, the mechanism mapping causes to each effect is autonomous and should not need to change when other mechanisms change [5, 26, 112, 114, 117]. This idea appears in multiple guises across the field: invariant causal prediction in classical causality, contrastive or auxiliary-variable nonlinear ICA, mechanism sparsity in CRL, modular decoders in deep SCMs, and environment-based objectives for robust learning. Although these formulations differ technically, they all try to overcome the same obstacle: the latent space of a deep model is too underconstrained unless one can exploit stable asymmetries in the data-generating process [5, 19, 61–80].

This survey is intentionally selective in one respect. The broader neighborhood of the field includes all of neural causal inference, invariant learning, causal reinforcement learning, causality for fairness, and generic disentanglement. Those topics are referenced where necessary, but the emphasis here is narrower: methods that either learn causal structure with deep models or learn latent variables endowed with causal semantics. That focus helps prevent the survey from collapsing into a general review of causality and deep learning.

Methods

Structural causal models, identifiability, and classical discovery foundations

Most work surveyed here can be read as extending the structural causal model tradition rather than replacing it. The SCM formalism, together with the Markov and faithfulness assumptions, translates causal questions into graph-theoretic, probabilistic, and functional conditions [1–5]. Classical discovery

methods exploited these conditions in three main ways. Constraint-based methods such as PC and FCI infer graph features from patterns of conditional independence, with FCI and RFCI explicitly permitting latent confounding and selection bias [2, 13–18]. Score-based methods search over DAGs to optimize a penalized likelihood or equivalent score, often under sparsity or decomposability assumptions [15, 20, 21]. Functional-model approaches identify directionality by imposing asymmetry on the data-generating mechanism, for example through non-Gaussianity in LiNGAM, independence between residuals and inputs in additive-noise models, or post-nonlinear restrictions [22–27].

These classical results remain critical for deep-learning methods because they define the irreducible identifiability barriers. For instance, with only observational data and unrestricted function classes, a DAG is generally identifiable only up to a Markov equivalence class. To go beyond that, methods require non-Gaussian noise, additive-noise structure, post-nonlinear form, multiple environments, interventions, or temporal order [19–27, 47–60]. Deep methods inherit exactly this logic. Neural parameterization increases expressivity, but it does not by itself create identifiability. If anything, it often intensifies non-identifiability unless paired with restrictions or richer data [6, 8–11].

A related lesson is that evaluating discovery in terms of graph recovery alone can obscure the distinction between statistical and causal identification. Papers such as NOTEARS, DAG-GNN, or DECI may optimize a consistent objective under certain assumptions, yet their outputs still depend sensitively on whether those assumptions are plausible for the data at hand. The criticism of over-reading DAG learners as “causal” is therefore well taken. The jump from graph structure to causal semantics always requires a substantive modeling commitment, not just a computational innovation [1, 2, 5, 9, 35].

Differentiable and neural causal discovery

The decisive computational shift in recent causal discovery was the replacement of combinatorial DAG search with continuous optimization. NOTEARS showed that acyclicity can be expressed by a smooth equality constraint on a weighted adjacency matrix, converting DAG learning into constrained continuous optimization [28]. This idea catalyzed a large family of methods. Nonlinear NOTEARS extended the objective to generalized additive or neural function classes [29]. DAG-GNN embedded the graph into a variational graph-neural-network architecture, enabling nonlinear and mixed-data modeling [30]. GraN-DAG used neural conditional distributions together with gradient-based optimization of a DAG constraint [31]. GOLEM simplified the objective for linear-Gaussian settings, while DAGMA proposed a more stable optimization route through a log-det acyclicity characterization [33, 34]. Taken together, these works made graph learning compatible with standard deep-learning toolchains and expanded the scale and flexibility of score-based discovery. ³

These methods offered three major benefits. First, they allowed rich conditional models without requiring handcrafted conditional-independence tests. Second, they aligned discovery with SGD, automatic differentiation, and amortization. Third, they made it natural to incorporate neural side modules for missingness, mixed data, or downstream causal queries. Yet they also exposed new weaknesses. The acyclicity penalty can be numerically delicate. Many objectives are nonconvex and sensitive to scaling, initialization, and regularization. The critique that some formulations prioritize sparse variance-explanation over genuine causal recovery is not merely philosophical; it points to objective misspecification under realistic deviations from model assumptions [32, 35].

A second branch emphasized interventions and posterior inference. DCDI extended differentiable discovery to interventional settings, explicitly modeling how interventions sharpen orientation and identifiability [36]. ENCO separated edge existence from edge orientation and exploited observational plus interventional data without requiring an acyclicity penalty during optimization [38]. DiBS reframed structure learning as variational inference over latent graph representations, yielding approximate

posterior distributions over graphs rather than single estimates [39]. Gradient-based posterior inference further developed this Bayesian perspective [40]. DECI moved still further, unifying discovery and effect estimation in a single flow-based additive-noise model capable of structure learning and causal query answering [43]. The common trend is clear: the field is moving from point-estimate graph search toward uncertainty-aware, intervention-aware, end-to-end causal modeling.

Another noteworthy line replaced local edge optimization with global generative search. GFlowCausal used generative flow networks to sample high-scoring graphs proportionally to reward [41], and DiffAN used diffusion-style modeling to search over topological orderings rather than directly over graphs [42]. These methods are valuable because the posterior over DAGs is often highly multimodal; amortized samplers can capture this better than single-solution optimizers. Even when they do not outperform simpler methods on all benchmarks, they expand the design space for discovery and point toward more realistic uncertainty quantification.

Finally, several neural discovery methods explore regularization beyond graph recovery. CASTLE couples auxiliary causal graph discovery to supervised prediction, using the estimated graph as an inductive bias for generalization [37]. This idea is theoretically and practically important because it treats discovery as a regularizer rather than an endpoint. In many applications, especially scientific discovery from modest sample sizes, one may care less about exact structure recovery than about whether causal inductive biases improve robustness and downstream intervention prediction.

Temporal, dynamical, and interventional discovery

Time series are a natural meeting point for causality and deep learning because temporal order partially alleviates orientation ambiguity. Nevertheless, temporal discovery is not a trivial extension of static discovery. Autocorrelation, latent lags, contemporaneous dependencies, regime changes, subsampling, and nonstationarity all complicate inference [10, 47–60]. Classical Granger causality formalized predictive precedence [47], and graphical time-series methods extended that logic to multivariate systems [48, 60]. Non-Gaussian structural VAR identification exploited asymmetries in innovations to orient edges [49].

Runge and collaborators substantially advanced modern causal discovery for time series through the PCMCi family, which combines conditional-independence testing with lag screening to scale to large, nonlinear multivariate systems while controlling false positives [50, 51]. These methods remain among the most influential for scientific applications because they are explicit about assumptions, lag structure, and uncertainty. They also underscore an important point often forgotten in deep-learning discussions: for many time-series problems, carefully designed statistical tests can still compete strongly with neural methods.

Deep learning enters more decisively when functional dependencies are nonlinear or high-dimensional. Neural Granger Causality used sparsity-structured MLPs and RNNs to estimate nonlinear Granger graphs [52]. TCDF used attention and temporal convolutions to detect lagged causal impact [53]. DYNOTEARS adapted differentiable acyclicity ideas to dynamic graphs [45]. These methods typically trade some statistical transparency for modeling flexibility, and their value is greatest when the state dimension is moderate but nonlinear relations are strong. For large-scale systems with heterogeneous or mixed-type signals, however, hybrid strategies that combine temporal order with stronger priors may still be preferable.

Subsampling is another recurrent challenge. When measurements occur more slowly than the causal process, naive discovery can invent spurious links or miss true ones. Constraint-optimization approaches addressed this in the classical literature [54], and more recent work leveraged proxy-variable ideas or limited two-slice assumptions to improve identifiability in subsampled settings [57, 58]. Likewise,

regime-dependent and nonstationary time series motivate discovery methods that treat environmental variation as a source of information rather than a nuisance [56]. Across these works, a general lesson emerges: richer data collection protocols often matter more than richer function classes. If temporal resolution, intervention labels, or context indices are poor, neural sophistication cannot rescue identifiability.

Causal representation learning through nonlinear ICA and identifiability theory

Among all methodological families in CRL, the most conceptually important may be the identifiable latent-variable program rooted in nonlinear ICA. Hyvärinen and Pajunen showed long ago that unconstrained nonlinear ICA is non-identifiable [61]. This impossibility result is directly analogous to the non-identifiability of generic latent spaces in deep generative models. The breakthrough came from introducing auxiliary structure. Time-contrastive learning exploited nonstationarity in time series [62]. Generalized contrastive learning used auxiliary variables more broadly [63]. The iVAE framework then unified nonlinear ICA and variational autoencoders, showing that conditional priors on latent variables can restore identifiability up to simple transformations [64]. Closely related flow-based and structured-noise extensions broadened the class of identifiable latent models [65, 66]. This body of work established a rigorous template for CRL: latent recovery becomes possible not from raw observations alone, but from the combination of a nonlinear observation model with structured variation across time, views, environments, or interventions. 4

This theory also recontextualized the disentanglement literature. Unsupervised disentanglement in generic VAEs is impossible without inductive bias or supervision, as argued forcefully by Locatello et al. [68]. Weak supervision, grouped samples, and carefully chosen augmentations can help [69], but the more profound lesson is that “disentanglement” is not a purely geometric property of a latent space; it depends on assumptions about the data-generating process. Causal representation learning sharpened that point by insisting that latent dimensions may themselves be causally dependent and therefore need not be statistically independent. The target is not always independent factors, but causally meaningful factors linked by an SCM [6, 11, 68–70].

Several papers made this transition explicit. Robustly disentangled causal mechanisms argued that latent factors should be validated by their interventional robustness rather than by axis alignment alone [70]. Zimmermann et al. connected contrastive learning to inversion of the data-generating process, showing how representation objectives can recover latent content under suitable sampling conditions [67]. More recent work unified many CRL settings under invariance principles, arguing that what is often identified is not causality per se but latent variables aligned with stable symmetries induced by temporal, environmental, or intervention structure [80]. This is a crucial conceptual refinement. It suggests that CRL is less a single algorithmic recipe than a family of identification strategies built on stable asymmetries in data.

Weak supervision, temporal structure, multiple views, and mechanism sparsity

Because pure observational recovery of latent causal variables is generally impossible, practical CRL methods exploit richer supervision in weak but carefully designed forms. Weakly supervised causal representation learning introduced paired samples before and after unknown interventions, together with implicit latent causal models, and showed that under mild assumptions causal factors and structure become identifiable from low-level observations [71]. Sparse-perturbation approaches studied similar settings where only a few latent mechanisms change at a time, linking representation recovery to sparse intervention structure [72]. These methods are attractive because they require less annotation than full labels while providing much stronger identification signals than static i.i.d. data.

Mechanism sparsity emerged as another influential principle. The central idea is that latent factors interact through sparse mechanisms over time or across interventions, and this sparsity itself can anchor the latent space. Lachapelle and colleagues showed that such sparsity can identify nonlinear latent variables up to permutation under appropriate graph-connectivity conditions and proposed VAE-style implementations with learnable masks [73–75]. This line is important for two reasons. First, it creates a direct bridge between causal discovery and representation learning: one learns both causal factorization and sparse relations among factors. Second, it moves beyond factor independence, allowing causally dependent latent variables as long as the mechanisms relating them are sparse and structured.

Temporal structure provides a different route. CITRIS considered image sequences together with intervention targets and proved identifiability for scalar and multidimensional causal factors in temporal settings [76]. Temporally disentangled representation learning studied latent process recovery from general temporal data [77]. The appeal of temporal CRL is that dynamics naturally separate slowly varying, intervention-sensitive, and causally linked components. Yet this advantage is fragile. If temporal correlations are dominated by nuisance dynamics or if interventions are unobserved, temporality alone may merely induce a dynamically useful but not causally clean latent space.

Multiple views and multiple environments broaden the strategy set further. Multi-view CRL with partial observability showed that jointly observed modalities can identify shared latent structure up to smooth bijections under graphical criteria [78]. Multi-domain CRL via weak distribution shifts used environmental variation as a substitute for more explicit supervision [79]. The invariance-principle perspective then argued that many of these settings can be unified: time, views, and environments each provide pockets of data in which some aspects of the latent structure remain stable while others vary [80, 81]. This unification is theoretically elegant and practically useful, because it frames CRL as a problem of choosing the right data heterogeneity for the latent variables one hopes to identify.

Deep generative causal models and the bridge between representation, discovery, and counterfactuals

A large and growing set of methods embeds causal assumptions directly into deep generative models. CausalVAE replaced the independent latent prior of standard VAEs with a causal layer transforming exogenous noises into causally related concepts [84]. DEAR and related weakly supervised generative causal models argued that independent priors are inappropriate when semantic factors are causally connected, and that generative models should reflect these dependencies explicitly [83, 85]. ICM-VAE pushed the principle of independent causal mechanisms into the encoder-decoder design [83]. These models are significant not only because they produce more interpretable latent traversals, but because they make counterfactual generation a first-class objective.

Parallel work developed deeper structural causal models for intervention and counterfactual reasoning. CEVAE used deep latent-variable modeling for causal effect estimation under hidden confounding [91]. Deep Structural Causal Models for Tractable Counterfactual Inference introduced a general framework combining SCM semantics with normalizing flows and variational inference, enabling abduction, intervention, and counterfactual queries in high-dimensional settings such as medical imaging [90]. DECI further merged discovery and inference in a flow-based additive-noise model [43]. These models illustrate one of the most promising directions in the field: replacing the old pipeline of “first estimate graph, then estimate effects” with integrated models that represent uncertainty jointly over structure, mechanisms, and latent states.

Causally informed GANs and related generative models form another strand. CausalGAN incorporated a causal graph into adversarial image generation [88]. Counterfactual Generative Networks decomposed image generation into independent causal mechanisms for shape, texture, and background and used

that decomposition to synthesize counterfactual images for more robust classification [87]. Functional causal models with generative neural networks used implicit generative mechanisms rather than restrictive parametric equations [89]. This line shows that causal modeling in deep learning is not limited to tabular causal discovery; it increasingly concerns controllable generation, mechanistic interpretability, and counterfactual data synthesis.

Finally, several works now study directly how discovery and representation should interact in perceptual domains. Discovery in physical systems from videos recovered causal relations among latent physical states from visual observations [97], while relational inductive biases such as neural relational inference [98] provided useful infrastructure for modeling dynamic interacting systems. The broader implication is that high-dimensional causal learning may require a three-layer decomposition: representation learning to recover compact state variables, discovery or mechanistic modeling over those variables, and causal reasoning modules for interventions and counterfactuals. Treating any one layer in isolation often leads either to non-identifiable latents or to brittle causal claims.

Evaluation protocols, datasets, and what current benchmarks really test

Evaluation remains one of the hardest problems in the field. For tabular discovery, standard metrics such as structural Hamming distance, skeleton F1, and edge-orientation accuracy are useful, but they often understate uncertainty and can reward near-equivalent graphs. For CRL the problem is worse, because the goal is not merely disentanglement but recovery of variables with causal semantics. Metrics based solely on latent-axis alignment or downstream prediction can miss the essence of intervention validity and mechanism modularity [6, 11, 80].

Benchmark design has therefore become a research topic in its own right. CausalWorld provided robotic tasks with explicit interventions over underlying factors [86]. Causal3DIdent and its temporal descendants, popularized by the CITRIS and contrastive-learning literatures, supplied visually rich settings with known latent causal graphs [67, 76]. More recent benchmarks such as CausalVerse and Causal3D aim to close the gap between toy synthetic worlds and realistic visual complexity by coupling configurable high-fidelity simulations with ground-truth latent causal processes [106, 107]. Meta-evaluations of CRL now increasingly focus on reproducibility, benchmark realism, and whether evaluation metrics actually detect intervention faithfulness rather than generic representation quality [108]. ⁵

A key lesson from recent benchmark work is that many current methods succeed only when the data-generation protocol is tightly aligned with their assumptions. Temporal methods benefit from clean intervention labels and relatively simple nuisance factors. Multi-view methods benefit from well-designed overlap among views. Sparse-mechanism methods benefit from true sparsity in latent transitions or intervention targets. This is not necessarily a weakness; it reflects the logic of identifiability. But it does mean that benchmark diversity is essential. A method that performs well on one synthetic family may be validating its assumptions more than its generality.

Challenges and Prospects

The first enduring challenge is identifiability under realistic observation models. Deep networks can represent extremely rich latent-variable transformations, but causal semantics do not emerge from expressivity alone. The field now broadly accepts that some form of auxiliary information is necessary: interventions, multiple views, temporal nonstationarity, environment shifts, grouped observations, sparse mechanism changes, or prior mechanistic structure [6, 11, 61–80]. The open question is not

whether such information is needed, but which forms are both theoretically sufficient and practically available in real scientific and industrial workflows.

The second challenge is the relationship between graph recovery and useful causal abstraction. In many perceptual domains, recovering the “true” low-level graph may be impossible or scientifically irrelevant. What matters instead may be discovering a higher-level causal state space that supports stable intervention prediction and compositional generalization. This suggests that CRL should sometimes be judged less by exact recovery of simulator latents and more by whether the learned abstraction is causally sufficient for downstream tasks. Recent work on invariance-based unification pushes in this direction, but formal criteria for good causal abstractions remain underdeveloped [6, 80, 81].

A third challenge is latent confounding, feedback, and cyclicity. Much of the deep discovery literature still assumes acyclic SCMs and no unobserved confounders, at least in the core model. Yet many real systems violate both assumptions. Classical methods such as FCI and RFCI offer partial solutions, but scalable deep analogues remain comparatively immature [17, 18]. The same applies to mechanisms with memory, delayed effects, and partially observed exogenous shocks in time series. Progress here will likely require more than better neural architectures; it will require new identification theory for latent, dynamic, and possibly cyclic causal representations.

A fourth challenge is uncertainty quantification. Many applications of discovery, especially in science and medicine, need calibrated posterior uncertainty over graphs, mechanisms, and counterfactuals. Point-estimate DAGs are often too brittle. Bayesian differentiable methods such as DiBS and gradient-based posterior inference are promising, but they are still computationally demanding and not yet the default for large-scale problems [39, 40]. A mature field will need approximate posteriors that are both faithful and operationally tractable.

A fifth challenge is external validity. Recent scoping and application-focused reviews emphasize that success on synthetic benchmarks does not guarantee performance in messy, real-world settings with measurement error, selection bias, or unclear intervention semantics [95, 96]. This criticism is especially sharp for differentiable DAG learners, which can look strong under benchmark conventions while being fragile under rescaling, misspecification, or weak signals [35]. Better grounding in real scientific workflows is therefore essential. ⁶

The prospects, however, are substantial. One promising direction is integrated discovery-representation systems that learn latent variables and graph structure jointly, rather than treating one as fixed input to the other. Another is active and interventional data collection: if the goal is identifiability, data should be acquired with that goal in mind. A third is multimodal causal learning, where partial observability across modalities provides exactly the kind of structured heterogeneity that CRL theory predicts is useful. A fourth is benchmark development: realistic simulators with explicit latent mechanisms can still play a major role, provided that evaluation focuses on intervention and counterfactual validity rather than cosmetic disentanglement. Finally, the increasing intersection of causality with foundation models may create new tools for weak supervision, scientific prior elicitation, and multimodal intervention design, but only if paired with disciplined causal evaluation rather than generic language-model competence.

Conclusion

Causal representation learning and causal discovery in deep learning should be understood as two sides of the same problem: how to move from high-dimensional observations and statistical regularities to variables and structures that support intervention, counterfactual reasoning, and transport across changing conditions. Classical causal discovery contributed the core identifiability logic and the SCM

language. Deep learning contributed expressive observation models, scalable optimization, and tools for high-dimensional data. The field’s most important achievements arise when these strengths are combined rather than when either side is treated as sufficient by itself.

Three lessons stand out from the literature. First, identifiability is the central organizing principle. Whether the auxiliary signal is time, interventions, views, or environments, successful methods succeed because they exploit asymmetries that generic latent-variable learning ignores. Second, graph discovery and representation learning should increasingly be studied jointly. Discovery without representation assumes away a central difficulty; representation without discovery risks learning latents with no stable causal meaning. Third, evaluation must become more causal in substance. Benchmarks and metrics should test modularity, intervention validity, uncertainty, and abstraction quality, not only reconstruction or disentanglement scores.

If these lessons are followed, the field can plausibly move from elegant proofs on synthetic data toward robust causal learning systems for perception, scientific discovery, medicine, and embodied intelligence. The most promising future models will likely not be “deep” or “causal” in isolation, but deliberately hybrid systems that use deep neural machinery to recover candidate variables, causal principles to constrain and test them, and interventional data to refine them.

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1 2 Towards Causal Representation Learning

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3 DAGs with NO TEARS: Continuous Optimization for Structure Learning

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4 Variational Autoencoders and Nonlinear ICA: A Unifying Framework

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5 <https://arxiv.org/abs/2510.14049>

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